The Architect of the Molecular Scissors: A Technical and Historical Dossier on Virginijus Šikšnys and the CRISPR Revolution

The trajectory of modern biotechnology was irrevocably altered in the early second decade of the twenty-first century, a period defined by the transition of the CRISPR-Cas system from an obscure bacterial immune mechanism to a universal tool for genomic surgery. At the center of this transformation stands Virginijus Šikšnys, a Lithuanian biochemist whose career serves as a bridge between the classical enzymology of the Soviet era and the synthetic biology of the future. This report provides an exhaustive collection of research materials, anchor scenes, and biochemical insights intended to facilitate a nuanced understanding of Šikšnys's contributions, the institutional barriers he navigated, and the scientific controversies that continue to shape his legacy.

A) ANCHOR SCENES

SCENE #1: The Mechanical Prehistory of Logic

The narrative of Virginijus Šikšnys does not begin in a high-tech laboratory but in the tactile, rhythmic environment of a post-war accounting office in Šiauliai, Lithuania. Born in 1956, Šikšnys was raised in an era of reconstruction, where the physical tools of logic were mechanical rather than electronic.

- **What happened:** A young Šikšnys is introduced to the world of numbers and physical mechanisms through his parents' profession. This early exposure to quantitative processes fostered a mindset that viewed complex systems as assemblies of moving parts that could be calculated and understood.
- **Date/Place:** Late 1950s/Early 1960s; Šiauliai, Lithuania.
- **Who was involved:** Virginijus Šikšnys; his father (a financial accountant/bookkeeper).
- **Concrete texture:** The setting is a sparse, professional office in a city still recovering from the devastation of World War II. Šikšnys recalls the "unique chance to explore mechanical calculators" used by his father to "crunch numbers".[1] These devices, with their metal gears and manual cranks, provided a physical representation of mathematical logic—a precursor to the "molecular scissors" he would later help define.
- **Why it mattered:** This scene establishes the foundation of Šikšnys's scientific identity as a "chemical enzymologist." His preference for kinetics and chemical mechanisms over purely descriptive biology can be traced back to this early fascination with the physical realization of logic.[1] It underscores the transition from a mechanical understanding of

math to a molecular understanding of the "arithmetic" of DNA.
- **Evidence quality:** Primary. Sourced from Šikšnys's own autobiographical reflections for the Kavli Prize.[1]
- **Myth risk:** Low. The anecdote is presented as a personal memory of childhood curiosity rather than a dramatic, destiny-altering epiphany.

SCENE #2: The Key to Independence

The transition from a curious child to a practitioner of science occurred in the chemistry laboratory of his secondary school, facilitated by a mentor who recognized that talent requires autonomy.

- **What happened:** Ms. Genovaite Jagminiene, a chemistry teacher, recognizes Šikšnys's aptitude and grants him unsupervised access to the school's laboratory, allowing him to venture beyond the standard Soviet curriculum.
- **Date/Place:** Early 1970s; Secondary school, Šiauliai, Lithuania.
- **Who was involved:** Virginijus Šikšnys; Ms. Genovaite Jagminiene.
- **Concrete texture:** The "key to the chemistry laboratory" is a physical and symbolic object in Šikšnys's memory.[2] The workspace was a modest school lab where he performed "simple experiments" and solved chemistry problems that were often "beyond the curriculum".[1] This period was punctuated by his participation in the "Chemistry Olympiads," a rigorous system of city and national competitions that demanded high-level problem-solving.[1]
- **Why it mattered:** This is the formative moment of scientific autonomy. The trust granted by Ms. Jagminiene allowed Šikšnys to develop the experimental instincts and self-reliance necessary for a researcher operating in a relatively isolated scientific community in Vilnius. It also highlights the importance of the Olympiad system in nurturing elite scientific talent in the Baltic region.[1]
- **Evidence quality:** Primary. This is a recurring theme in his interviews and official biographies.[1]
- **Myth risk:** Low/Medium. While the "key to the lab" is a classic trope in scientific biographies, Šikšnys provides specific names and contexts (the Olympiads) that ground the story in historical reality.

SCENE #3: The Moscow Synthesis

In 1978, Šikšnys moved from the familiar environment of Vilnius University to the Department of Chemical Enzymology at Lomonosov Moscow State University, an institution that represented the pinnacle of Soviet scientific research.

- **What happened:** Šikšnys enters a PhD program where he transitions from organic chemistry to biochemistry. He joins a cohort of scientists who apply the rigorous principles of chemical kinetics to biological systems, specifically studying the thermoinactivation of proteolytic enzymes.
- **Date/Place:** 1978–1983; Lomonosov Moscow State University, USSR.
- **Who was involved:** Virginijus Šikšnys; Karel Martinek (PhD Advisor).

- **Concrete texture:** Moscow State University was "probably the best place for PhD studies in the former Soviet Union" at the time.[1] The atmosphere was one of intense quantitative rigor. Šikšnys describes a culture where chemists brought their "rigorous understanding of the kinetics and mechanisms of chemical reactions into biochemistry".[1] His work focused on the "mechanisms of thermoinactivation of proteolytic enzymes".[1]
- **Why it mattered:** This scene defines his intellectual methodology. He did not approach CRISPR as a "biologist" looking for an immune system, but as a "chemical enzymologist" looking for a catalytic mechanism. This perspective was crucial for his later ability to demonstrate the programmable nature of Cas9 *in vitro*.[1]
- **Evidence quality:** Primary. Detailed in his autobiography and confirmed by institutional records.[1]
- **Myth risk:** Low. The details align with the historical reputation of the Moscow school of enzymology.

SCENE #4: The April Rejection and the CRISPR Scoop

The most dramatic turning point in Šikšnys's career occurred in the spring of 2012, characterized by a series of publication delays that would eventually cost him a share of the Nobel Prize.

- **What happened:** Having successfully demonstrated that Cas9 is a programmable DNA-cutting enzyme, Šikšnys submits his manuscript to *Cell Reports*. The paper is rejected without review, forcing him to move to *PNAS*, where a slow peer-review process allows a competing team (Doudna and Charpentier) to publish their findings first.
- **Date/Place:** April 18, 2012 (Submission to *Cell Reports*); May 21, 2012 (Submission to *PNAS*); Vilnius, Lithuania.
- **Who was involved:** Virginijus Šikšnys; Giedrius Gasiūnas (PhD student and lead author).
- **Concrete texture:** The scene is one of quiet scientific triumph followed by institutional frustration. The Vilnius team had already achieved the "Eureka" moment of seeing Cas9 cut DNA at a site of their own choosing.[4] However, the rejection from *Cell Reports* was a "desk rejection," meaning it was never even sent to peer reviewers.[3] While their paper languished in the *PNAS* review process for months, the Doudna/Charpentier paper was submitted to *Science* on June 8, fast-tracked, and published online by June 27.[5]
- **Why it mattered:** This scene highlights the "prestige game" in science and the barriers faced by researchers in "less mainstream institutes".[6] It is the focal point of the controversy regarding priority in the CRISPR field. While Šikšnys technically described the system's programmability in a submitted manuscript first, the publication lag meant he lost the "race" for priority in the public eye.[8]
- **Evidence quality:** High. Publication submission and acceptance dates are publicly verifiable and documented by science historians.[3]
- **Myth risk:** Medium. The narrative often frames this as a "snub" based on geography; while this may be true, the technical differences between the papers (such as the single-guide RNA engineering) also played a role in the reviewers' decisions.

SCENE #5: The Martinsried Connection

A lesser-known but critical scene occurred in 1993, which laid the groundwork for Šikšnys's expertise in the structural biology of DNA-protein interactions.

- **What happened:** Šikšnys travels to the Max Planck Institute for Biochemistry in Germany as a visiting scientist in the laboratory of Nobel Laureate Robert Huber.
- **Date/Place:** 1993; Martinsried, Germany.
- **Who was involved:** Virginijus Šikšnys; Robert Huber.
- **Concrete texture:** This was a period of transition for Lithuania, which had recently restored its independence. International collaboration was a new and vital frontier. In Huber's lab, Šikšnys immersed himself in "crystallographic studies of restriction enzymes".[1] The workspace was a high-intensity environment focused on solving the three-dimensional structures of proteins.
- **Why it mattered:** This experience transformed his lab from a group focused on kinetics to one capable of high-level structural biology. Without this mastery of X-ray crystallography and structural analysis, his team would not have been able to provide the mechanistic insights into Cas9 that characterized their 2012 paper.[1]
- **Evidence quality:** Primary. Documented in his professional biography and institutional history.[3]
- **Myth risk:** Low.

SCENE #6: The Validation in Oslo

The final scene in this narrative arc provides a resolution to the priority disputes, showcasing Šikšnys as a recognized equal in the CRISPR trio.

- **What happened:** Šikšnys stands on a stage in Oslo to receive the 2018 Kavli Prize in Nanoscience, sharing the honor with Jennifer Doudna and Emmanuelle Charpentier.
- **Date/Place:** September 4, 2018; Oslo, Norway.
- **Who was involved:** Virginijus Šikšnys; Jennifer Doudna; Emmanuelle Charpentier.
- **Concrete texture:** The ceremony is an elite gathering of the scientific world. Šikšnys receives a gold medal and a share of the 1 million dollar prize.[13] The phone call from the president of the Norwegian Academy of Science and Letters had come as a "surprise," representing a moment of "international recognition" that transcended the Nobel selection.[13]
- **Why it mattered:** This scene serves as the "Myth Resolution." While the Nobel Prize is limited to three recipients and famously excluded Šikšnys in 2020, the Kavli Prize serves as a definitive statement from the global scientific community that his independent discovery was equal in significance to the work done at Berkeley and the Max Planck Institute.[8]
- **Evidence quality:** Primary. Well-documented by the Kavli Foundation and international press.[12]
- **Myth risk:** Low.

B) DIRECT QUOTES FROM THE SCIENTIST

Quote	Context/Date	Source
"Imagine that this sentence is our DNA and words are our genes. Now as you see there is a long letter here. Using these molecular scissors we can replace the wrong letter in a word or even replace the wrong word with the right word. Even more we can edit the whole sentence."	2018 Kavli Prize Interview explaining the technology to a general audience.	[15]
"Actually, CRISPR was discovered as an anti-viral defense system in bacteria. But there are many others, including one called restriction-modification system... I'm sure that further studies of bacteria will lead to new discoveries."	2018 Kavli Panel Discussion on the origins of the field and its future.	[2]
"My path to science was similar to Jennifer's because I had a really great chemistry teacher in high school. She boosted my interest in chemistry. She trusted me with the key to the chemistry laboratory where I was allowed to do some simple experiments."	2018 Panel Discussion reflecting on early mentorship and independence.	[1]
"I'd like to cut through all the mysteries that surround DNA you never know when you struck your luck again."	2018 Video Interview regarding his ongoing motivation and research.	[15]
"The potential and rapid development of life sciences captivated my attention... bringing with them a rigorous understanding of the kinetics and mechanisms of chemical	Autobiography describing his transition from organic chemistry in Moscow.	[1]

reactions into biochemistry."		
"My scientific career would never have been the same without the students and my lab team. I always had the privilege to work with brilliant students that not only did excellent science in the lab, but also created a fantastic atmosphere."	Autobiography reflecting on the collaborative nature of the Vilnius lab.	[1]
"My wife Dangira who also has a PhD in chemistry was often the first who heard about new discoveries and failures in the lab."	Autobiographical note on his family's scientific background.	[1]

C) HUMAN REALITY DETAILS

The following details provide a nuanced view of Virginijus Šikšnys as a practitioner of science, a mentor, and a citizen of a rapidly developing scientific landscape.

- **Upbringing in Šiauliai:** Born in 1956, Šikšnys grew up in a city where the post-WWII environment was characterized by financial bookkeeping and mechanical logic, which influenced his later preference for quantitative "numbers-based" biochemistry.[1]
- **Chemistry Olympiad Success:** His early competitive streak was nurtured through the Lithuanian system of chemistry competitions, which required solving complex problems outside the standard school curriculum.[1]
- **Academic Heritage:** His PhD advisor in Moscow, Karel Martinek, was from Czechoslovakia, highlighting the international but Soviet-centered academic networks of the late 1970s.[1]
- **Domestic Scientific Partner:** His wife, Dangira, is not just a spouse but a fellow PhD in chemistry, serving as his primary sounding board for laboratory data and theoretical challenges.[1]
- **Mentoring Style:** Šikšnys is credited with creating a "fantastic atmosphere" in the lab, where students like Giedrius Gasiūnas were given the lead on groundbreaking projects, demonstrating a lack of scientific ego in favor of team success.[1]
- **Career Loyalty:** Despite his global reputation and the opportunity to move to prestigious Western institutes, Šikšnys has spent nearly his entire career in Vilnius, contributing significantly to the fact that the biotech sector now accounts for 2.5% of Lithuania's GDP.[11]
- **Structural Focus:** He is a master of X-ray crystallography, having solved approximately 15 out of 50 available restriction endonuclease tertiary structures in collaboration with international teams.[3]

- **Humor and Perspective:** He jokingly refers to the "epigenetic footprint" he may have left on his daughter Jurga, who chose a career in management despite being raised in a house of two PhD chemists.[1]
- **Institutional Leadership:** Since 2007, he has chaired the Institute of Biotechnology Council, steering the research direction of Lithuania's primary life science hub through the transition into the European Research Area.[3]
- **The "Desk Rejection" Resilience:** Despite the 2012 *Cell* rejection, Šikšnys did not abandon the work; he immediately pivoted to *PNAS*, demonstrating a quiet persistence in the face of institutional gatekeeping.[3]
- **Breadth of Interests:** While synonymous with CRISPR today, he worked for over two decades on restriction endonucleases, proving his expertise in the broader field of protein-DNA interactions before the CRISPR "fad" began.[1]
- **Founding Entrepreneur:** He is a co-founder of CasZyme, a company designed to commercialize CRISPR discoveries, showing a commitment to the practical application of his fundamental research.[20]

D) SCIENCE CONNECTION (Intro Biochemistry)

Best-fit Topic: Enzymes and Nucleic Acid Metabolism

Virginijus Šikšnys's work is a quintessential study in **enzyme specificity**, **catalysis**, and **protein-DNA binding**. It connects directly to the core principles of how enzymes recognize substrates and the role of cofactors (in this case, RNA) in directing catalytic activity.

Precise Contribution

Šikšnys independently demonstrated that the Cas9 protein is an RNA-guided endonuclease that can be programmed to cleave any specific DNA sequence *in vitro*, thereby proving the feasibility of Cas9 as a universal genome-editing tool.[1]

Bridge Explanation for Intro Students

In intro biochemistry, we learn that enzymes have active sites that fit specific substrates. Cas9 is unique because its "active site" for DNA recognition isn't fixed in the protein's structure; instead, it uses a strand of RNA as a "search term." By providing Cas9 with a different RNA sequence, we can change which DNA sequence it binds to and cuts. Šikšnys's team was the first to show that this system could be taken out of a bacterium and used in a test tube to cut a DNA molecule exactly where the researchers wanted. This proved that Cas9 wasn't just a bacterial defense—it was a programmable machine for molecular surgery.[1]

Scientist's Own Analogy

Šikšnys uses the **"Molecular Scissors"** analogy. He describes DNA as a long "sentence" and genes as the "words." Cas9 acts as scissors that can find a specific "wrong letter" and cut it out, allowing the cell to replace it with the "right letter".[13]

E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

The story of Virginijus Šikšnys is inextricably linked to the sociology of credit and the politics of prestige in modern science.

Item	Priority and the 2020 Nobel Prize Snub
What's agreed	Šikšnys submitted his findings on programmable Cas9 to *Cell Reports* on April 18, 2012, while Doudna and Charpentier submitted their paper to *Science* on June 8, 2012. Both teams independently arrived at the conclusion that Cas9 could be programmed with RNA.[3]
What's disputed	The Nobel Committee awarded the 2020 prize only to Doudna and Charpentier. The dispute centers on whether Šikšnys's exclusion was a matter of technical merit (the Doudna team's engineering of the single-guide RNA) or institutional bias against a "less mainstream" university in Lithuania. Some argue Šikšnys's paper was "incomplete" because he did not emphasize the tracrRNA requirement as clearly as the Berkeley team, while others point out that his paper acknowledges tracrRNA multiple times and that the "single-guide" RNA is an engineering step, not a fundamental discovery.[5]
Best Sources	*Forbes* analysis "Prof. V. Siksnys deserves to share the glory" [8]; Reddit Biochemistry debate on the "tracrRNA" vs "sgRNA" distinction.[5]

Item	The Role of Marketing and Prestige
What's agreed	The Doudna and Charpentier paper was published in *Science* within two weeks of submission, while Šikšnys's paper was rejected by *Cell* without review and took four months to be published in *PNAS*.[5]
What's disputed	Whether Šikšnys "failed to effectively communicate the importance or urgency" of his discovery, or if the scientific publishing system is biased toward elite American institutions.[6] There is a significant discussion in

	the CRISPR community about whether Šikšnys was "snubbed" because he didn't know how to "play the prestige publication game".[7]
Best Sources	*Nature* snapshot of Lithuanian life sciences [3]; Reddit Chemistry "Why didn't Virginijus Šikšnys win the Nobel Prize?".[7]

Item	**The tracrRNA Necessity**
What's agreed	Both crRNA and tracrRNA are required for the natural Cas9 system to function. The Doudna/Charpentier team linked them into a single-guide RNA (sgRNA).[21]
What's disputed	Some critics claim Šikšnys "didn't find the importance of both crRNAs," making his initial findings less impactful for genome editing in human cells. However, proponents of Šikšnys note that his *PNAS* paper acknowledges the tracrRNA component and that the fundamental discovery was the *programmability* of the protein itself.[5]
Best Sources	*PNAS* 2012 paper [3]; *Science* 2012 paper.[4]

F) “NOT ON WIKIPEDIA” GOLD

These items represent deep-dive findings from institutional reports, academic memoirs, and niche publications.

1. **The "MOBILI" Transformation:** A 2011-2012 Vilnius University report reveals that much of the infrastructure used for the CRISPR discovery was funded by the "EC FP7 program project MOBILI." This project was specifically designed to integrate Lithuanian biotechnology into the European Research Area and allowed Šikšnys to bring back seven experienced scientists who had been working abroad, fundamentally upgrading the lab's capabilities just before the CRISPR breakthrough.[19]
2. **Cas12m "Roadblock" Discovery:** While Wikipedia focuses on Cas9, Šikšnys’s lab recently (2023) published in *Nature* about "Cas12m" effectors. These proteins provide immunity not by cutting DNA, but by acting as a **physical roadblock** that inhibits transcription and replication. This represents a new paradigm in CRISPR function—interference without cleavage.[22]
3. **The Bicyclo[3.3.1]nonane Phase:** Before becoming a biochemist, Šikšnys’s undergraduate research was in pure organic synthetic chemistry, specifically the "stereochemistry of bicyclo[3.3.1]nonanes." This period in the stereochemistry lab of Prof. Petras Kadziauskas is where he first mastered the "mechanisms of chemical reactions" that he would later apply to enzymes.[1]

4. **TnpB and the "Minimal Core":** Šikšnys's team recently solved the structure of TnpB, which they identified as the "minimal functional core" of the entire Cas12 nuclease family. This research suggests that CRISPR-Cas systems evolved from much smaller, ancestral transposon-encoded proteins, providing a "molecular fossil" record of gene-editing evolution.[22]
5. **Magnetic Tweezers and Kinetic Proof:** To understand how Cas12f1 recognizes DNA, Šikšnys's lab utilized "single-molecule magnetic tweezers" measurements. This high-level biophysical technique allowed them to see the formation of 18 bp R-loops in real-time, a level of detail that goes far beyond standard biochemical assays.[22]

H) FLAGGED "MYTH-RISK" DETAILS

- **Claim:** Šikšnys sent his draft manuscript to Jennifer Doudna for review, and she "stole" the idea.
- **Where it appears:** Reddit Biochemistry threads and some sensationalist comments on Science news sites.[5]
- **Why sourcing is weak:** There is no primary evidence (emails, archival records) of this exchange. While scientists often share drafts with colleagues, the "theft" narrative is likely an exaggeration of the fact that both teams were presenting at the same conferences in early 2012.
- **Best available evidence:** The Doudna and Šikšnys papers are technically distinct (sgRNA engineering vs. *in vitro* Cas9 demonstration). Most experts believe they were independent, parallel discoveries.[4]
- **Claim:** The 2012 *Cell* rejection was a purely political act because the authors were from Lithuania.
- **Where it appears:** Lithuanian national pride articles and "Forbes" opinion pieces.[8]
- **Why sourcing is weak:** Desk rejections are extremely common for many reasons (formatting, perceived scope, competing papers already in the pipeline). While the *Cell* board later admitted the paper had great value [8], attributing the rejection solely to "anti-Lithuanian bias" is a narrative interpretation that cannot be definitively proven with available records.

LENS RECOMMENDATION

Lens Scoring Rubric

Lens	Score (0-3)	Justification
Eureka	2	He had a clear moment of seeing Cas9 cut DNA on command, but the "discovery" is overshadowed by the

		publication race.
Failure	3	The *Cell* rejection is a specific, well-documented setback that changed the entire trajectory of his public recognition and the Nobel outcome.
Mentor	2	The relationship with Ms. Jagminiene and his own role in fostering Gasiūnas's career are clear, but less dramatic than the priority dispute.
Controversy	3	The dispute over the "snub," the institutional bias in science, and the "marketing of ideas" offers a high-stakes narrative with multiple well-sourced perspectives.
Human Detail	2	The mechanical calculators and accounting background provide a unique "hidden" personhood, but are secondary to the scientific drama.
Last Experiment	1	Late-career moments are about awards and administration rather than a single dramatic closure scene.

Primary Recommended Lens: Controversy (The "Hidden Pioneer" Narrative)

Why it fits: The evidence overwhelmingly supports a story about the "prestige game" in global science. This lens allows the writer to use Scene #4 (The Rejection) as a visceral opening to explore why a brilliant scientist from a "small" country was overlooked. It connects the biochemistry of Cas9 directly to the social structures of science.

- **Best opening scene:** Scene #4 (The April 18, 2012, submission and subsequent rejection from *Cell*).
- **One quote to use early:** Quote #1 (The "sentence/word" DNA analogy), used to show how clearly he understood the technology he was about to lose credit for.
- **Cautions:** Be careful to avoid a "victim" narrative. Šikšnys himself is often humble and team-oriented; the controversy should be framed as a systemic issue in science rather than a personal grievance.

Backup Lens: Failure (The "Scooped" Narrative)

Why it fits: Within 1 point of the top score, this lens focuses on the "near miss." It is a scene-driven narrative about the agonizing months between April and September 2012.

- **Best opening scene:** Scene #4.
- **One quote to use early:** "I'd like to cut through all the mysteries... you never know when you struck your luck again".[15]
- **Cautions:** This lens can be overly dramatic. Ensure that the technical reasons for the Doudna/Charpentier success (the sgRNA engineering) are included for scientific accuracy.

CONCLUDING ANALYSIS: SOCIOLOGICAL AND BIOCHEMICAL SYNTHESIS

The case of Virginijus Šikšnys provides a profound second-order insight into the **mechanisms of scientific credit**. While the "Great Man" theory of history often searches for a single point of origin, the CRISPR revolution was a multi-nodal phenomenon. Šikšnys's career represents a high-signal instance of **institutional lag**—where the quality of the research (Lithuania) outpaced the scientific community's ability to recognize it.

His transition from the mechanical calculators of Šiauliai to the crystallographic studies of Martinsried suggests that his contribution to CRISPR was not a "lucky break" but the logical conclusion of a thirty-year obsession with the **physicality of information**. For Šikšnys, DNA was never just a biological code; it was a physical substrate that could be manipulated with the precision of a master clockmaker. This is evident in his lab's continued work on Cas12m "roadblocks" and TnpB "molecular fossils."

The 2.5% of GDP that biotechnology now represents in Lithuania [17] is perhaps the most enduring testament to his work, far more than the Nobel Prize that eluded him. He chose to build a world-class institute in a country that was, at the start of his career, scientifically isolated. This decision—to remain in Vilnius—is the ultimate "Not on Wikipedia" gold: the choice of a scientist who valued the **building of a scientific culture** over the pursuit of individual prestige in the West.

For students of BIOL 3030, Šikšnys's story is a reminder that enzymes are not just abstract concepts in a textbook; they are the tools of a global, high-stakes competition. His work on the **Type II CRISPR-Cas system** demonstrates that fundamental research into "obscure" bacterial defense mechanisms can lead to the most revolutionary technological advancements of a generation. The "molecular scissors" he helped build are now in the hands of thousands of researchers worldwide, a legacy that transcends the specific dates of a journal submission or the selection of a committee in Stockholm.

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